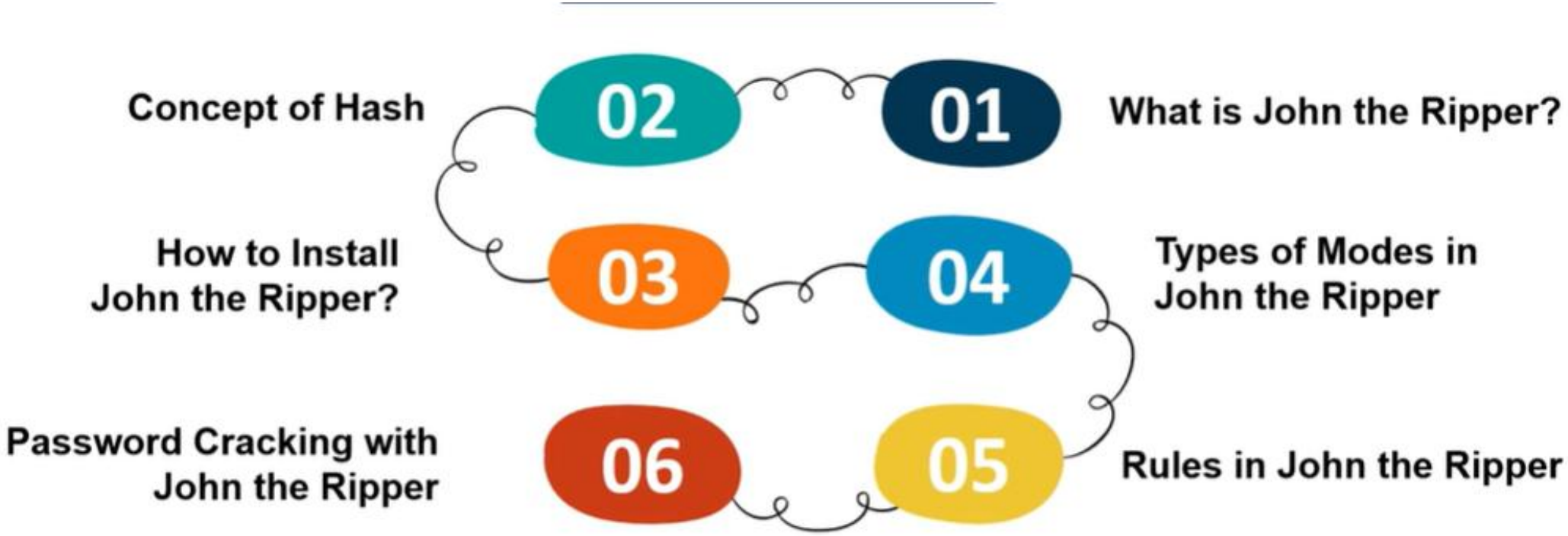
Password Cracking

TheJohnRipper

Contents


Concept of Hash
02
01
What is John the Ripper?
How to Install John the Ripper?
03
04
Types of Modes in John the Ripper
Password Cracking with John the Ripper
06
05
Rules in John the Ripper

What is John the Ripper?



John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.

- **Unix Flavors, macOS, Windows**
- **Web Apps**
- **Groupware**

- **Network Traffic Captures**
- **Encrypted Private Keys**
- **Filesystems & Disks**
- **Archives**

What is John the Ripper?

John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.



Security Experts



System

OR



File



Recover Passwords

What is John the Ripper?

John the Ripper is an Open-Source **Password Security Auditing** and **Password Recovery** Tool available for many **Operating Systems**.



Security Experts



Strength of Passwords

Strong Passwords

+

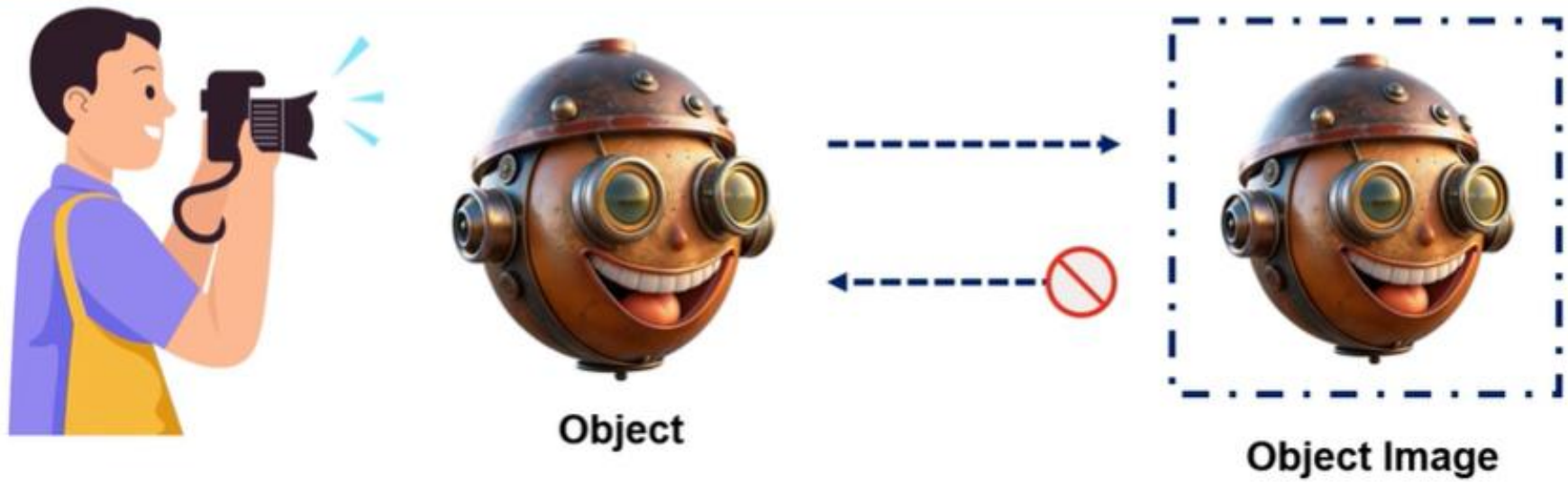
Weak Passwords



Improve Security

Concept of Hash

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



Why use Hash for Password Storage?

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



Why use Hash for Password Storage?

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



Hacker

482c811da5d5b4bc6d497ffa98491e38

MD5 Hash



Recover Passwords

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.

MD5 (Message-Digest Algorithm 5)



MD5

482c811da5d5b4bc6d497ffa98491e38

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.

SHA-256 (Secure Hash Algorithm 256-bit)



SHA-256 → **ef92b778ba27b969d9656c6d29f4a5e89e36dfeeb252c56b275becc77c6b2759**

Commonly Used Hash Types

Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.



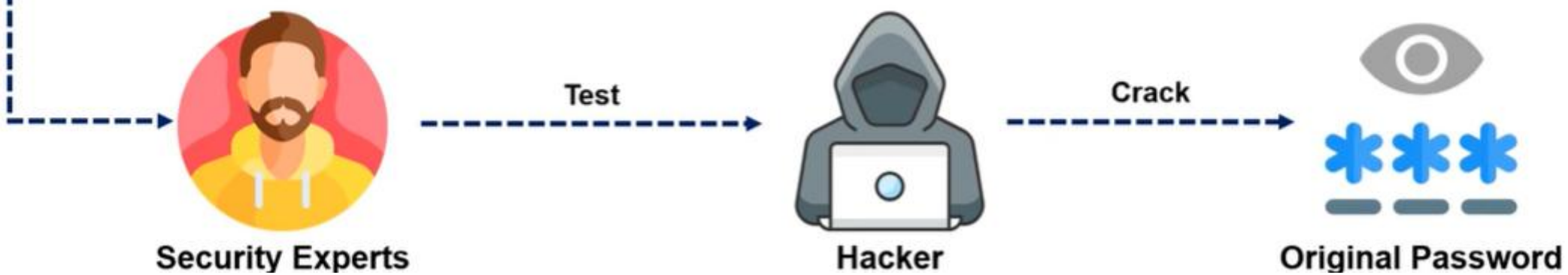
bcrypt

bcrypt

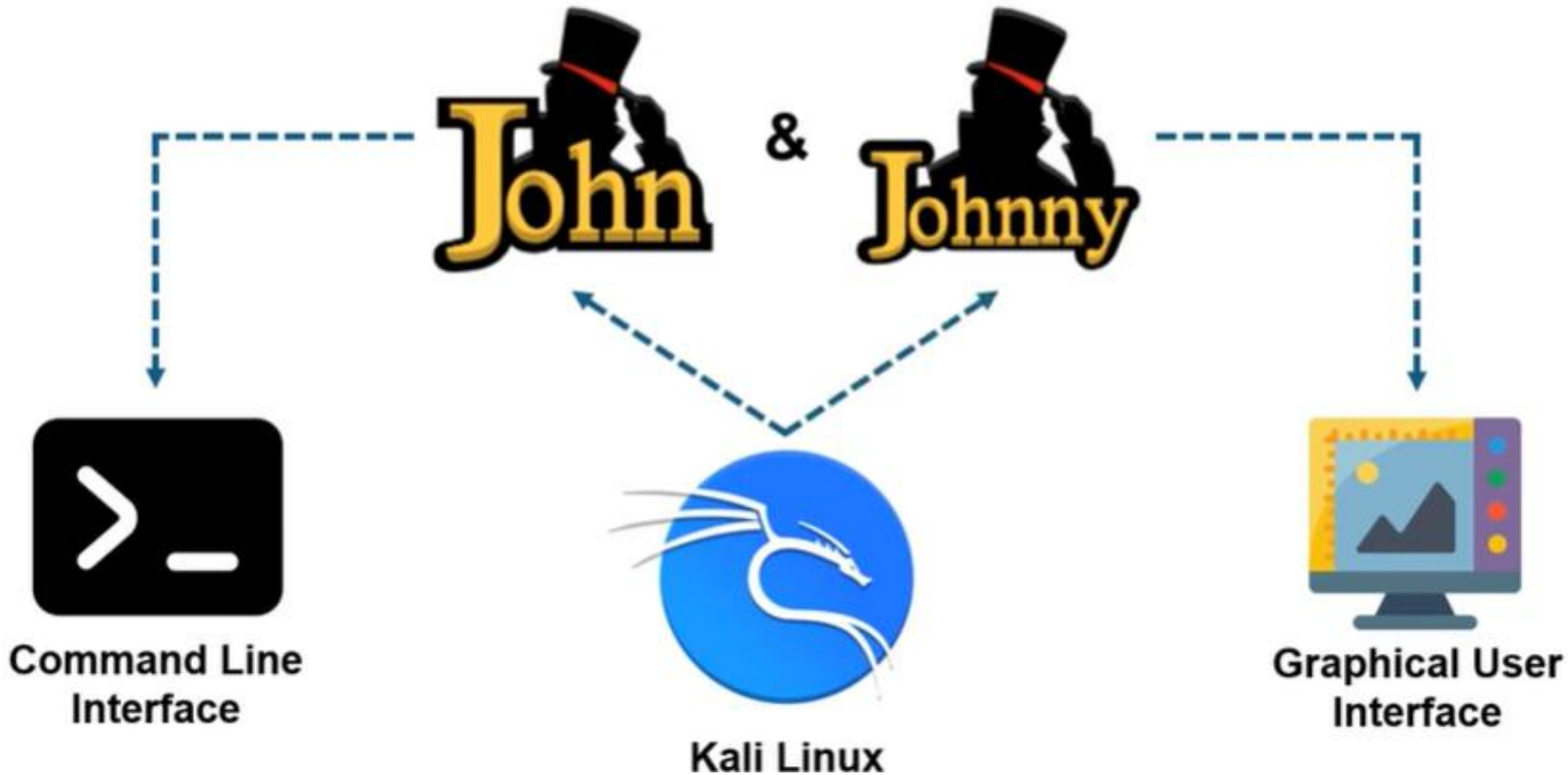
$2a$12$WXOeT6ezofQwXZRCMljM5ehfMQ1lgfjyhh/yFuSJwtaLWsdJI

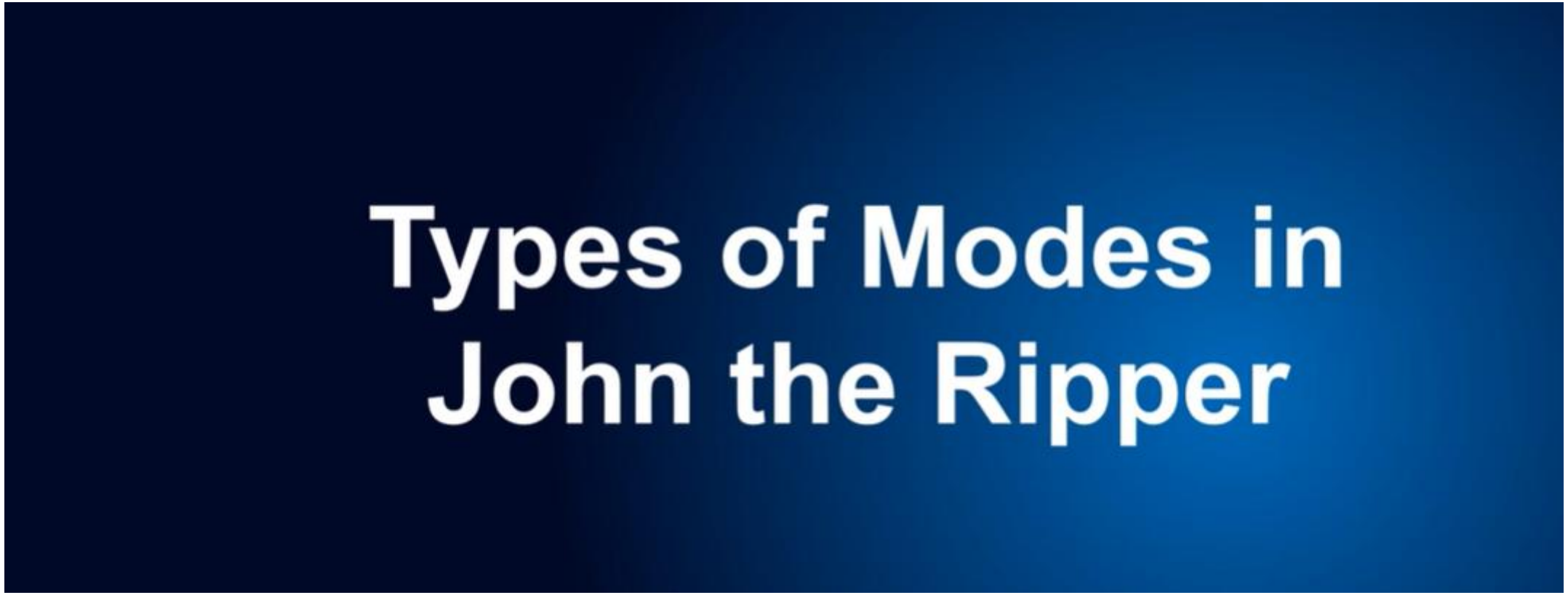
Why Cracking Hashes Important for Security Testing?

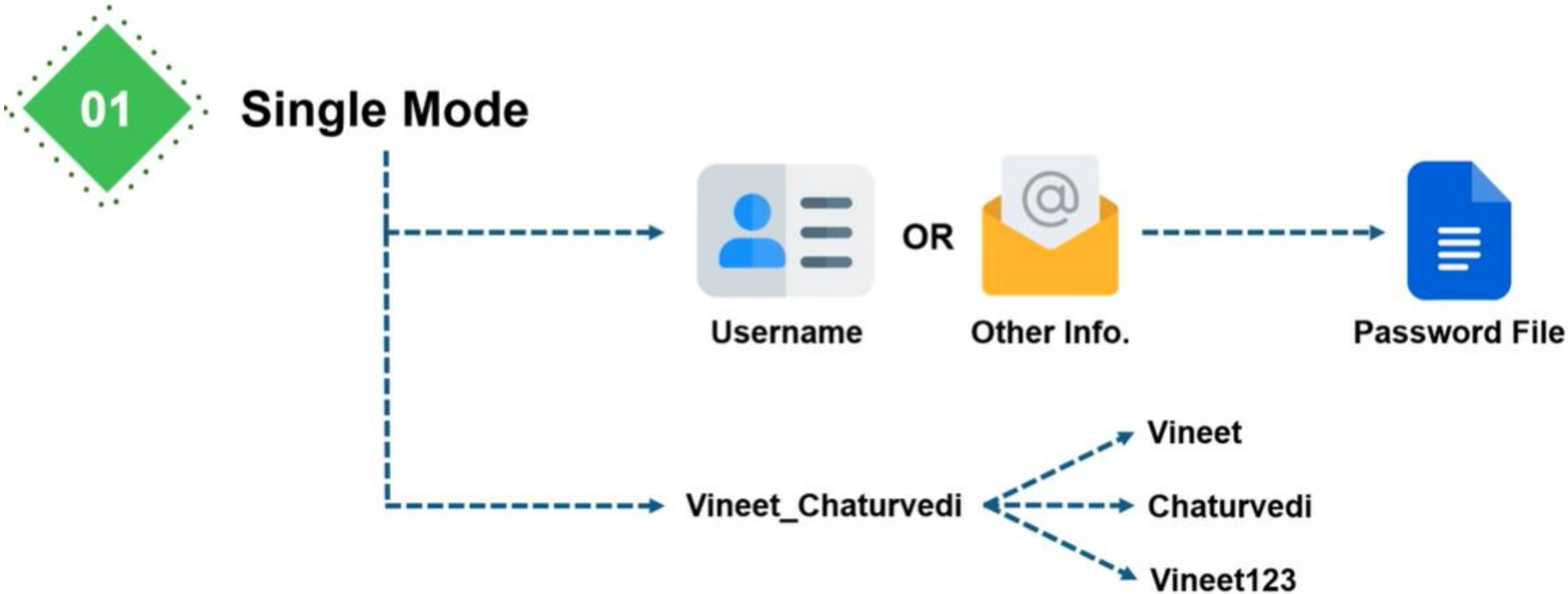
Hashing is a **Cryptographic** process that converts **Input Data** of any size into a **Fixed-Size String** of characters, known as a **Hash Value**.

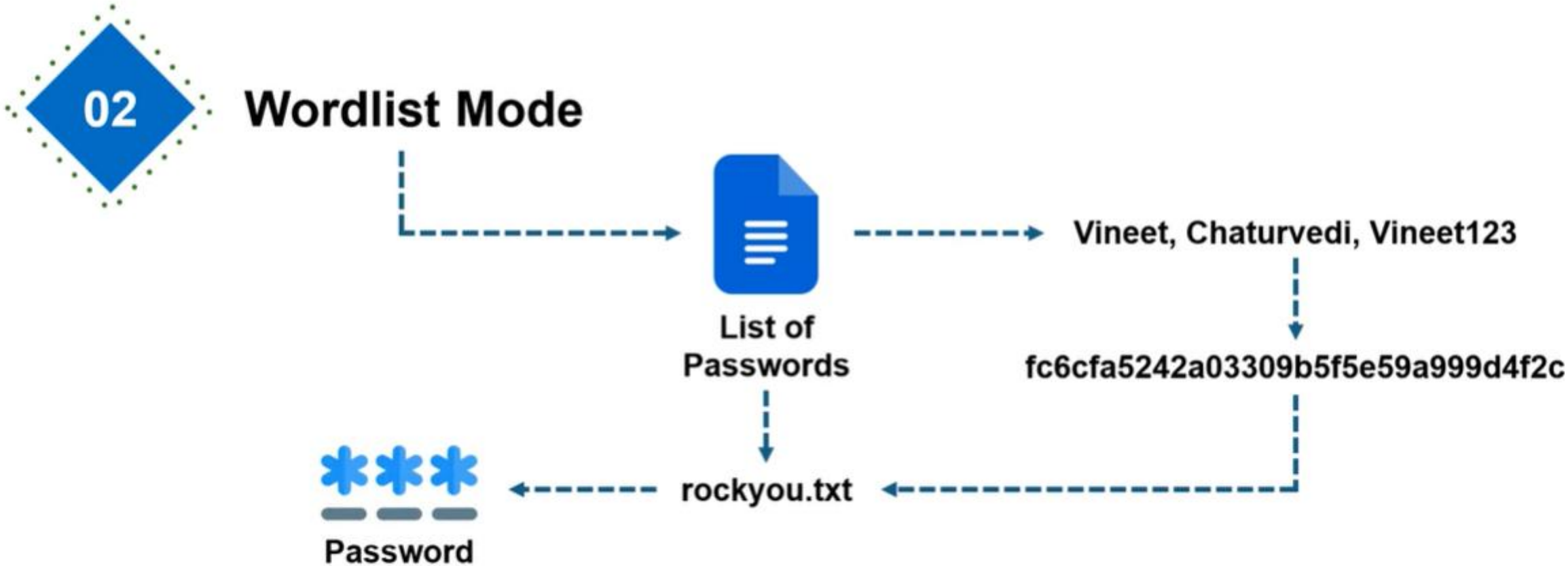


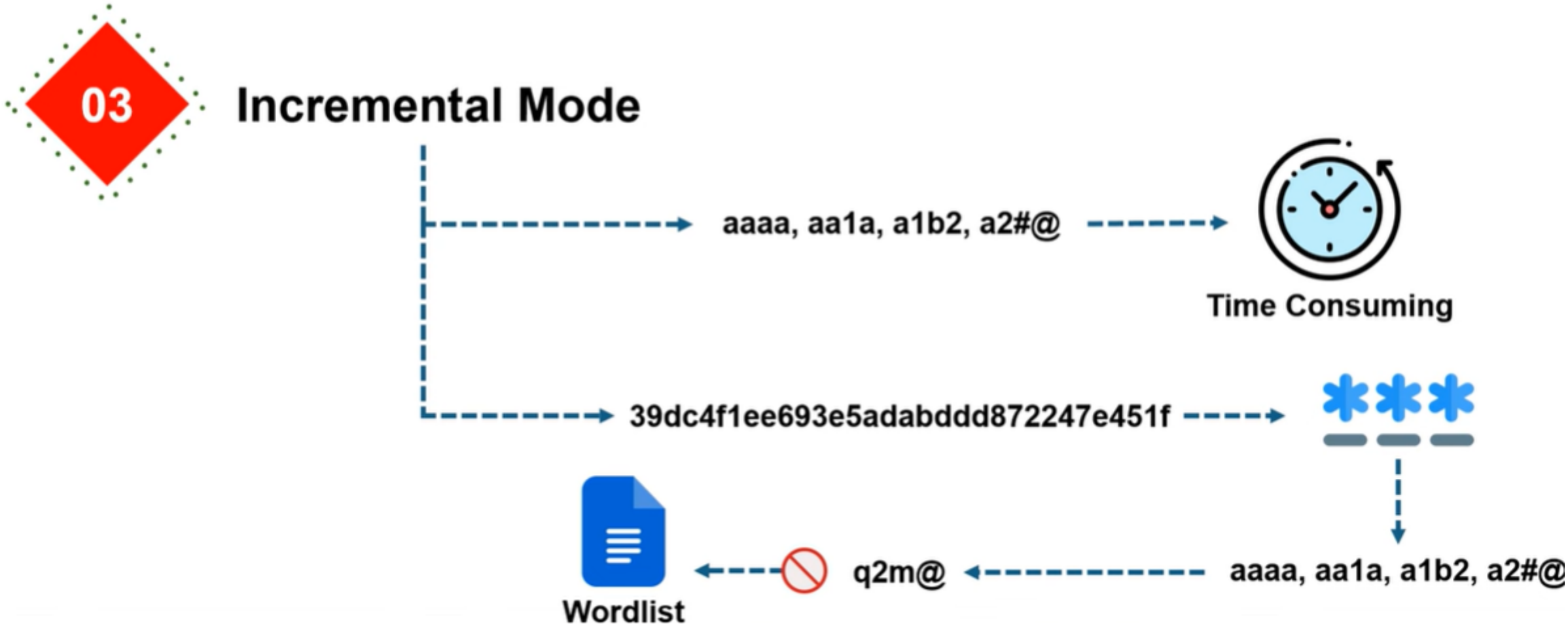
How to Install John the Ripper?




Types of Modes in
John the Ripper

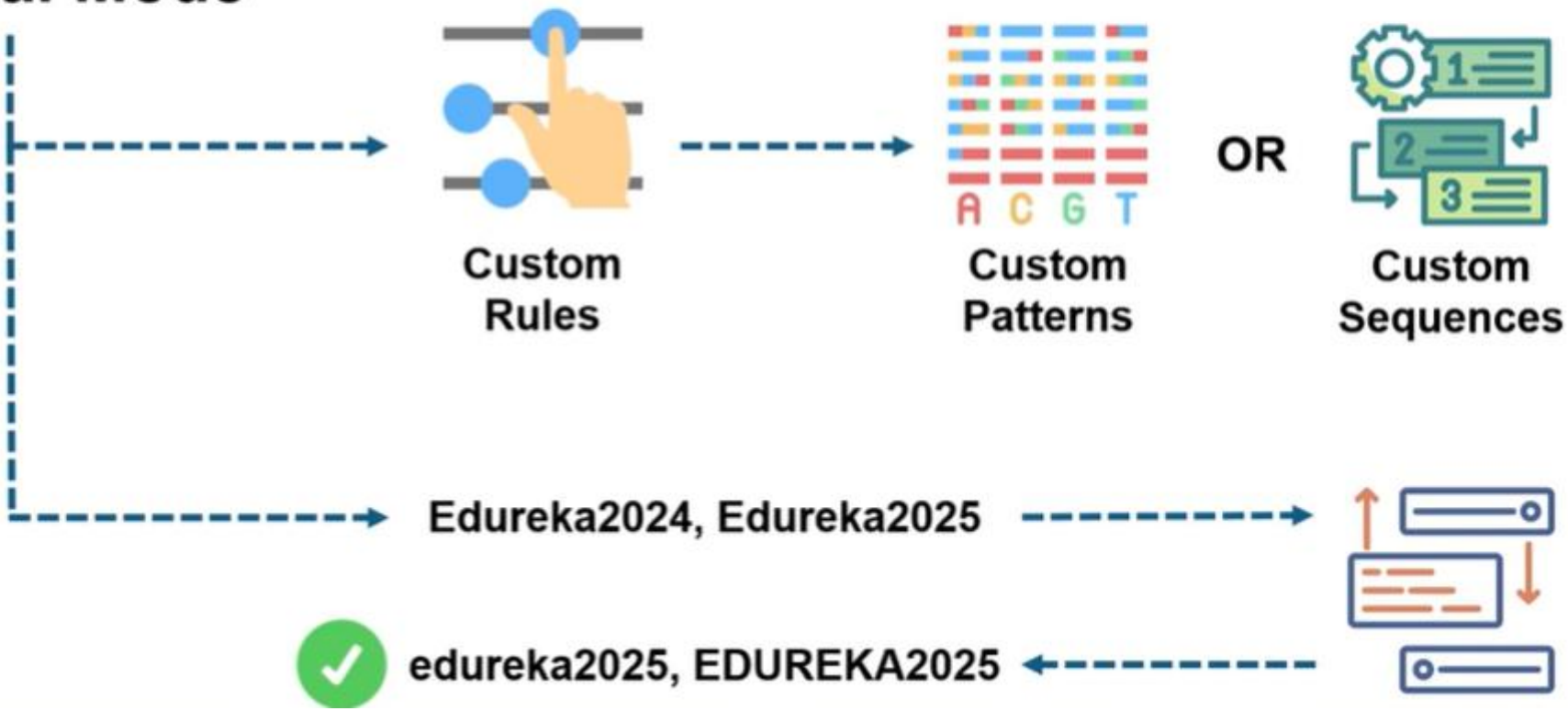

01
Single Mode
Username
OR
Other Info.
Password File
Vineet_Chaturvedi
Vineet
Chaturvedi
Vineet123


02
Wordlist Mode
List of
Passwords
Vineet, Chaturvedi, Vineet123
fc6cfa5242a03309b5f5e59a999d4f2c
rockyou.txt
Password

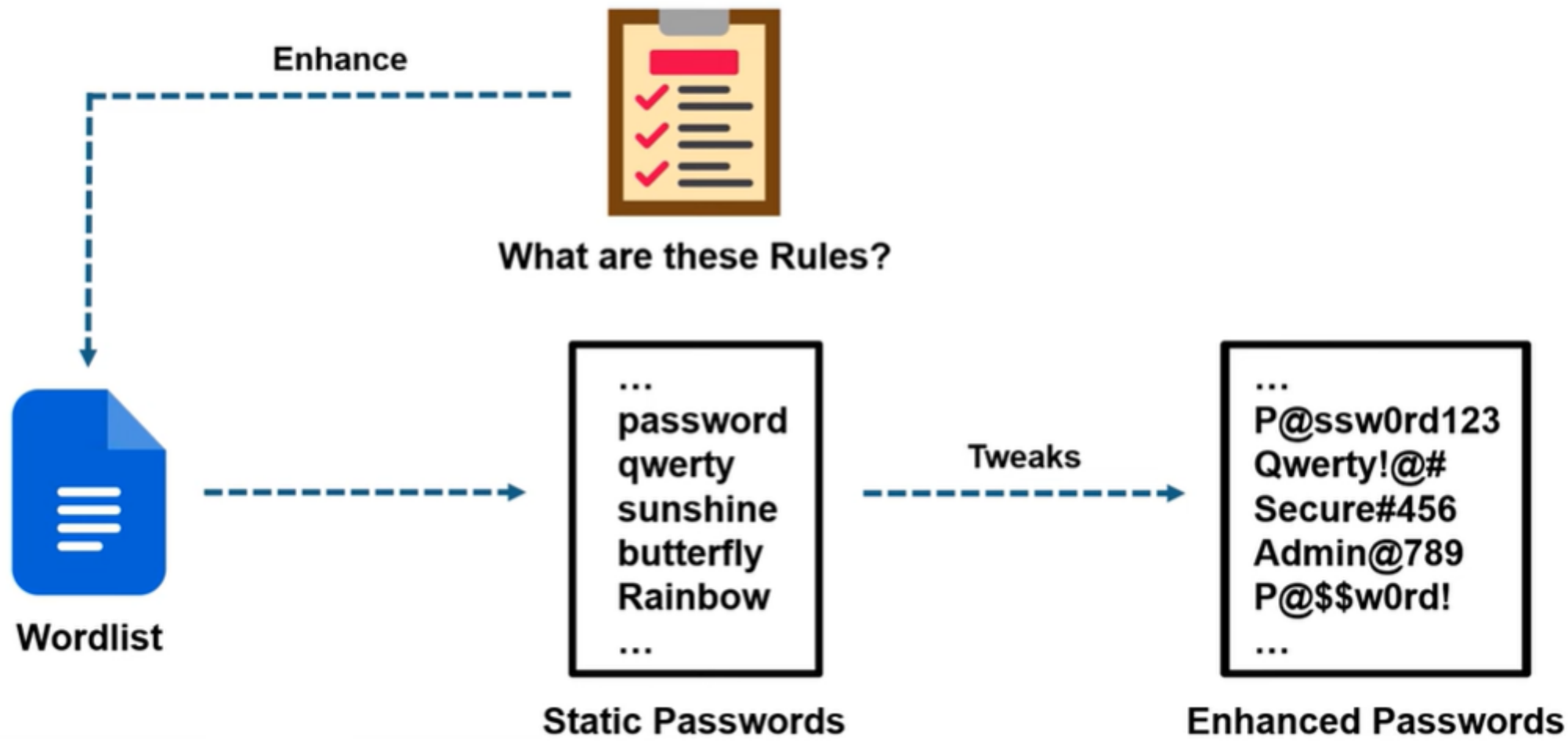

03
Incremental Mode
aaaa, aa1a, a1b2, a2#@
Time Consuming
39dc4f1ee693e5adabddd872247e451f
aaaa, aa1a, a1b2, a2#@
q2m@
Wordlist

04

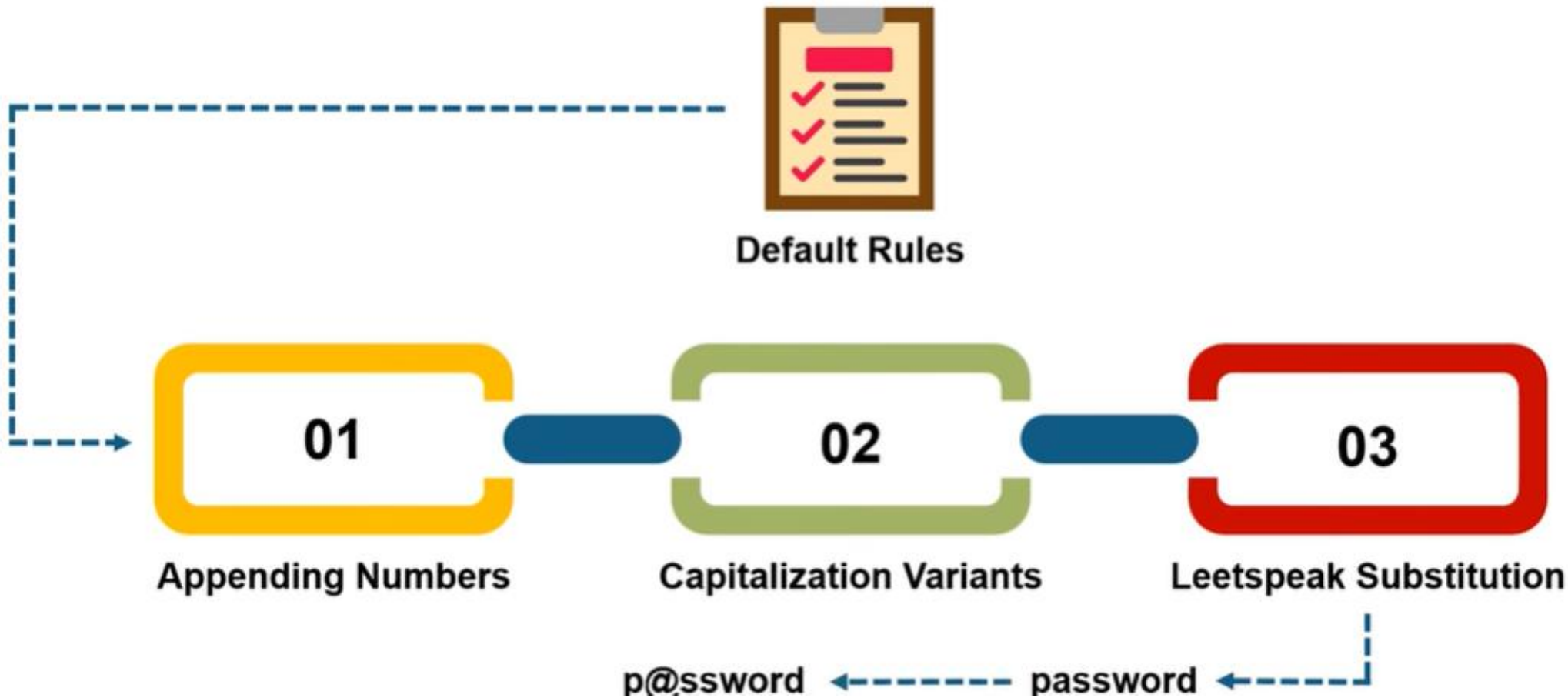
External Mode



Rules in John the Ripper



Rules in John the Ripper



Customizing Rules

Installing the John

- Checking
- john
- installing
- sudo apt install john
- Showing
- apt show john
- JOHN --HELP
- **GUI:**
- **sudoapt install johnny**
- **johnny**

Rules

- Sudo su-super file
- cd /etc/john-john directory
- nano john.confi

Search-list.Rules

[[A-a],[0-9]]

Why are Rules Important?



Rules



Birth Year/Common Suffix



Uppercase/Lowercase



Password Policy

Password Cracking with John the Ripper

- Create a hash file-touch hashfile.txt
- Wordlist file-touch wordlist.txt
- Locate file-cd /usr/share/wordlists
- Unzipfile-sudo gzip –d rockyou.txt.gz

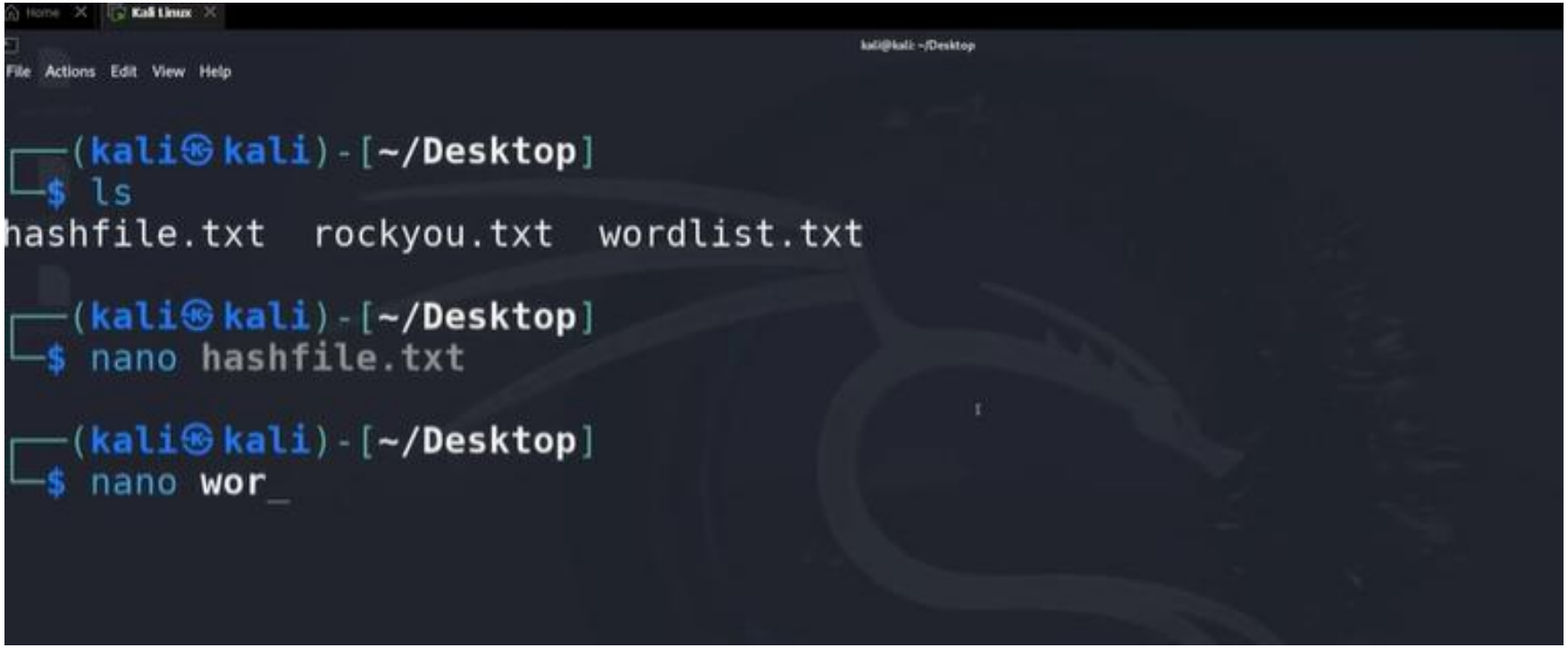
Supported Formats

- John –list=formats
- Basic syntax
- John hashfile.txt –format=RAW-MD5

Cracking Passwords using Single Mode

- Ls
- Cat hashfile.txt




Home
Kali Linux
kali@kali: ~/Desktop
File Actions Edit View Help
(kali㉿kali)-[~/Desktop]
$ ls
hashfile.txt rockyou.txt wordlist.txt
(kali㉿kali)-[~/Desktop]
$ nano hashfile.txt
(kali㉿kali)-[~/Desktop]
$ nano wor

Cracking Passwords using Wordlist Mode

- John –wordlist=wordlist.txt hashfile.txt –format=RAW-MD5
- John –show hashfile.txt –formate=RAW-MD5
- Updated hashvalues-before move into incremental mode
- Ls
- Nano hash.txt
- Shift toggle rule
- Ls
- Nano wordlist.txt
- John –rules=

This is an Offline Password Hash Cracking process using John the Ripper

- **sudo su -**
- Switches to the **root** account (superuser). The - makes the shell a *login* shell so root's environment and home directory (/root) are used.
- You were then at the root㉿kali prompt.
- **cat /etc/passwd**
- Prints the content of /etc/passwd. Each line describes one user account in this format:
- username:password_placeholder:UID:GID:GECOS:home_directory:shell

Example line from your output:
root:x:0:0:root:/root:/usr/bin/zsh

- root — username
- x — password placeholder (actual password hashes live in /etc/shadow)
- 0 — UID (0 = root)
- 0 — GID (group id)
- root — GECOS (full name/comment)
- /root — home directory
- /usr/bin/zsh — login shell
- /etc/passwd is world-readable so many commands can map uid→username.